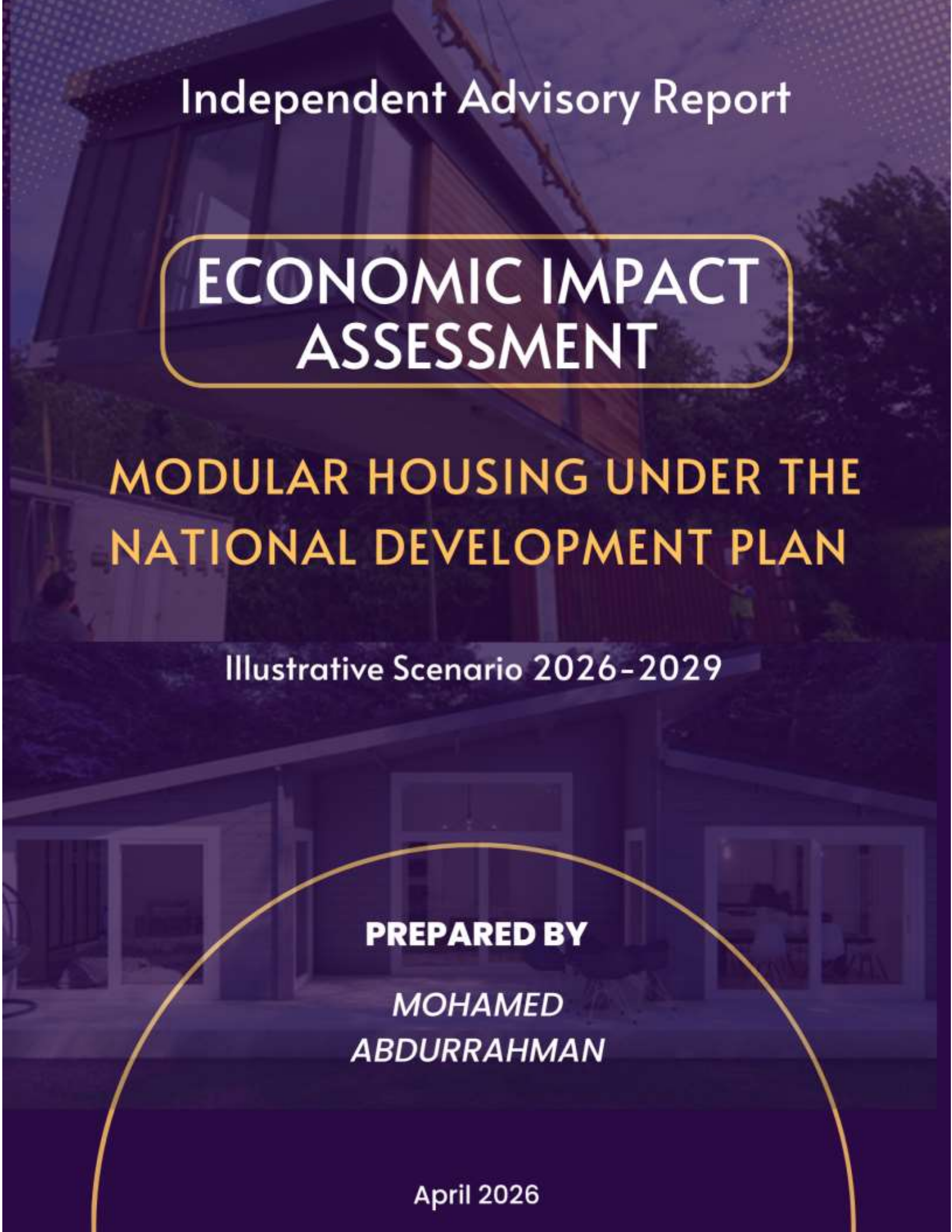




Independent Advisory Report

ECONOMIC IMPACT ASSESSMENT

MODULAR HOUSING UNDER THE NATIONAL DEVELOPMENT PLAN

Illustrative Scenario 2026-2029

PREPARED BY

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DISCLAIMER

This analysis was prepared independently for portfolio and educational purposes. It is not affiliated with any organisation. All data is from publicly available sources as of April 2026. Modular planning exemption is proposed and pending Cabinet approval; actual timing is uncertain. Rent-a-room applicability to garden units remains under discussion. All calculations are based on the accompanying Excel model.

Version: 1.0 – Economic Impact Assessment

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1. EXECUTIVE SUMMARY

This Economic Impact Assessment evaluates Ireland's modular housing delivery under the National Development Plan (NDP), incorporating verified April 2026 data on completions, costs, VAT rates, corporation tax incentives, and pending policy decisions.

Under the revised National Development Plan (July 2025), the Government allocated €102.4 billion for the 2026-2030 period, including €35.9 billion for housing and critical infrastructure. A binding target requires that at least 25% of all new social and affordable homes be delivered using Modern Methods of Construction (MMC), including modular systems. This analysis models the 2026-2029 period as an illustrative delivery window within this five-year capital framework.

Several key policy measures shape the current landscape. First, the Government has proposed exempting modular garden units up to 45 square metres from standard planning permission, provided at least 25 square metres of garden space remains. Second, Budget 2026 reduced VAT on new apartment sales from 13.5% to 9% until 31 December 2030. Third, the Finance Bill 2025 introduced a 125% corporation tax deduction (up to €50,000 per unit) for apartment blocks of 10 or more units, also expiring on 31 December 2030. Fourth, the rent-a-room scheme permits homeowners to earn €14,000 tax-free annually. However, coalition disagreement on whether garden units qualify for this relief has delayed the planning exemption.

Core Economic Concepts Applied:

Concept	Definition	Application
Gross Value Added (GVA)	Measure of economic output contributed by a sector, excluding taxes and subsidies	Derived from CSO construction sector averages (38% of gross delivery cost)
Exchequer Returns	Tax revenue generated across multiple streams	VAT (9% for apartments, 13.5% for houses), income tax, corporation tax (15% for large firms, with 125% deduction for apartment blocks of 10+ units)
Deadweight Loss	What would have happened anyway under traditional construction	Modular vs traditional counterfactual isolates incremental impact
Counterfactual Analysis	Baseline scenario without modular intervention	Traditional construction methods continue

Fiscal Multiplier Effects	Indirect employment and supply chain impacts	Estimated using international benchmarks (1.2-1.3x) due to absence of recent Irish input-output tables
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Key Metrics – Base Case (April 2026):

Metric	Value	Source
Total homes target (2030)	303,000	Delivering Homes plan (Nov 2025)
2025 completions (actual)	36,248	ESRI Spring 2026
2026 completions (forecast)	36,500	ESRI – mid-30,000s plateau
2027 completions (forecast)	38,000	ESRI – conditional on VAT measures
Modular cost saving vs traditional	15-20%	Industry benchmark
Modular time saving	50% (18→9 months)	Industry benchmark
VAT (houses)	13.5%	Standard rate
VAT (apartments – new)	9%	Budget 2026 (until 31 Dec 2030)
Corporation tax (large firms >€750m)	15%	OECD agreement effective 2024
Corp tax deduction (apartments 10+ units)	125% (max €50k/unit)	Finance Bill 2025
GVA multiplier (construction)	38%	CSO construction sector averages
Discount rate	4%	Irish Government Infrastructure Guidelines
Modular planning exemption	Proposed – pending Cabinet approval	Delayed due to rent-a-room disagreement
Rent-a-room annual cap	€14,000 (tax-free)	Revenue Commissioners

Key Results from Excel Model:

Metric	Value
Total homes delivered (2025-2029)	191,248
Total investment	€70.1 billion
Total Gross Value Added (GVA)	€7.0 billion
Total direct jobs supported	119,774 person-years
Total indirect jobs supported	179,661 person-years
Total jobs supported	299,435 person-years
Total exchequer return (tax)	€10.8 billion
Net Present Value (5-year)	€74.7 billion
Discount rate applied	4%

Conclusion: Modular construction offers significant efficiency gains (50% faster, 15-20% cheaper, 66% less waste) but remains dependent on pending policy decisions and resolution of severe labour shortages.

2. POLICY CONTEXT & IRISH MARKET OVERVIEW

2.1 National Development Plan & Housing Targets

The Irish Government's *Delivering Homes, Building Communities* plan (November 2025) establishes a national target of 303,000 new homes by the end of 2030, requiring an average delivery rate of approximately 50,500 homes per year. This target represents a significant acceleration from historical completion rates and reflects the Government's recognition of the housing crisis as a national priority under the National Development Plan (NDP), which allocated €275 billion for infrastructure investment including housing, transport, and energy.

Current Performance and Target:

Year	Completions	Target (50,500)	Source
2025	36,248	-14,252 (-28%)	ESRI Spring 2026
2026 (forecast)	36,500	-14,000 (-28%)	ESRI
2027 (forecast)	38,000	-12,500 (-25%)	ESRI conditional on VAT measures

The ESRI Spring 2026 Commentary notes that completions increased by 20.4% in 2025, driven largely by apartments in Dublin, but the rate of new construction commencements softened in late 2025, indicating a potential plateau in the mid-30,000s for 2026 and 2027. This plateau reflects structural constraints: labour shortages, supply chain bottlenecks, and planning system capacity.

Economic Insight: The gap between target and actual completions represents a structural supply deficit with material implications for house prices, rents, and inflation. Each 10,000-home shortfall is estimated to add approximately 3-5% to national average house prices, assuming constant demand. This means the 14,000 annual shortfall in 2025-2026 likely contributed 4-7% to house price inflation beyond normal market dynamics.

2.2 Modular Construction – Definition and Irish Context

Modern Methods of Construction (MMC), including modular construction, involve the off-site manufacture of building components under factory-controlled conditions, followed by rapid on-site assembly. For the Irish market, modular construction offers specific advantages that address critical bottlenecks in housing delivery:

Advantage	Irish Context
Speed	9 months vs 18-24 months traditional – critical for urgent housing targets
Labour efficiency	Reduces on-site labour requirements by up to 50% – critical given severe construction labour shortages (unemployment rate 4.3-4.7%, sector at "near full capacity")
Quality control	Factory conditions ensure consistent standards, reducing defects and long-term maintenance costs
Waste reduction	On-site waste reduced from 15% to 5% – aligns with circular economy and carbon reduction goals
Weather independence	Off-site manufacture mitigates Irish weather-related construction delays, which typically add 10-15% to traditional project timelines

The adoption of modular methods in Ireland has been slower than in the UK or Scandinavia due to historical policy uncertainty and limited manufacturing capacity. However, the Government's 2025 commitment to MMC under the Housing for All strategy has accelerated interest, with several modular factories now in development in the Greater Dublin Area and Midlands.

2.3 Modular Approval – Critical Policy Update (April 2026)

Current Status: Proposed – Pending Cabinet Approval

Event	Status	Source
Proposed regulations	Announced 21 April 2026 for 45 sq/m standalone modular homes	https://www.gov.ie/en/department-of-housing-local-government-and-heritage/press-releases/government-announces-changes-to-planning-exemptions-regulations/
Purpose	Exempt certain modular homes from standard planning permission	gov.ie
Garden unit rules	Exemption capped at 25 sq/m garden space remaining	gov.ie <i>Planning and Development (Exempted Development) (No. 4) Regulations 2026.</i>
Delay reason	Coalition disagreement on whether garden units qualify for rent-a-room scheme (€14,000 tax-free)	Business Post, 27 April 2026
Current status	Did not go to Cabinet as expected; "back and forth" ongoing	Business Post, 27 April 2026
Expected resolution	May return to Cabinet in coming weeks; unlikely to be scrapped entirely	Independent.ie (28 April 2026)

Economic Insight: The delay imposes a real economic cost. Each month of policy uncertainty delays modular project starts, pushing completion into 2027-2028 and delaying associated GVA and exchequer returns. Using a 4% discount rate, a 6-month delay reduces the present value of a €100 million modular project by approximately €2 million. For the full national programme (€70 billion over 5 years), each month of delay costs approximately €230 million in present value terms.

2.4 Construction Labour Market – Critical Constraint

Indicator	Value	Source
Construction employment (Q4 2025)	~175,000	CSO
Unemployment rate (2026 forecast)	4.3% – 4.7%	ESRI
Labour shortage severity	"Tens of thousands of additional workers needed"	ESRI December 2025
Sector capacity	"Near full capacity"	Central Bank February 2026
Wage inflation (construction, 2025)	~5-7%	Construction Industry Federation

Economic Insight: Labour is the single biggest brake on housing delivery. Even with modular construction reducing on-site labour requirements by up to 50%, the absolute number of available workers remains a binding constraint. The construction workforce would need to expand by approximately 25% (from 175,000 to 220,000) to meet the 50,500 annual target using traditional methods. Modular methods reduce this required expansion to approximately 12-15%, but still require significant recruitment and training. This suggests that meeting government targets requires either (a) significant labour supply expansion via apprenticeship programmes and targeted immigration, or (b) a sustained shift toward modular methods that use labour more efficiently – or ideally both.

2.5 Fiscal Incentives – Budget 2026 Measures

The October 2025 Budget introduced two significant incentives for apartment construction, specifically designed to address the housing crisis in urban areas:

Measure	Detail	Sunset	Source
VAT reduction	Reduced from 13.5% to 9% for new apartment sales and construction services	31 December 2030	Budget 2026

Corporation tax deduction	125% deduction (max €50,000 per unit) for apartment blocks of 10+ units	31 December 2030	Finance Bill 2025
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Economic Insight: These measures specifically target apartment construction, not houses. The rationale is economic fundamentals: apartments deliver higher density per unit of land (3-5x more units per hectare), making them more efficient in high-demand urban areas (Dublin, Cork, Galway) where land prices are highest. The 9% VAT rate reduces final purchase prices by approximately 4-5%, improving affordability for first-time buyers. The 125% deduction effectively subsidises apartment construction at a rate of approximately €10,000-€12,500 per unit (assuming 12.5% effective corporation tax rate on the 25% excess deduction). For a 100-unit apartment block, this represents a tax saving of €1.0-1.25 million, significantly improving project viability.

2.6 Rent-A-Room Scheme – Legal & Economic Implications

Aspect	Detail	Source
Annual tax-free limit	€14,000 per year	Revenue Commissioners
Current applicability	Homeowner's primary residence	Revenue Commissioners
Proposed extension	Garden modular units – under discussion	Business Post, 27 April 2026
Legal distinction	Licensee (not tenant) – fewer protections	Residential Tenancies Act
Key difference	Licensees lack security of tenure, rent pressure zone limits, and RTB dispute access	Residential Tenancies Act

Economic Insight: The rent-a-room distinction materially affects the economics of garden modular units:

Scenario	Annual After-Tax Return	Tenant Protections
Included (rent-a-room)	€14,000 (tax-free)	Licensee (limited)

Excluded (standard rental)	~€12,000 (after 20% tax + PRSI + USC)	Tenant (full RTA protections)
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The policy disagreement centres on this trade-off: inclusion boosts homeowner supply incentives (by €2,000-3,000 annually) but reduces tenant protections; exclusion protects tenants but reduces the financial case for homeowners to build garden units. The resolution of this disagreement is critical to unlocking the estimated 50,000-100,000 potential garden units that could be delivered on existing residential properties.

3. METHODOLOGY & DATA SOURCES

3.1 Economic Impact Framework

This assessment follows standard economic impact methodology for infrastructure projects, consistent with the Irish Government Infrastructure Guidelines (December 2023) and international best practice (HM Treasury Green Book, EC Guide to Cost-Benefit Analysis). The framework is designed to isolate the incremental impact of modular construction relative to the counterfactual of traditional building methods.

Core Economic Concepts Applied:

Concept	Definition	Application in This Assessment
Gross Value Added (GVA)	Measure of economic output contributed by a sector, excluding taxes and subsidies	Derived from CSO construction sector averages (38% of gross delivery cost)
Direct Effects	Immediate economic activity from construction spending	Jobs, GVA, tax revenue from building modular homes
Indirect Effects	Supply chain activity (materials, transport, professional services)	Estimated using 1.2-1.3 multiplier (international benchmark; caveated)
Induced Effects	Spending by construction workers in the wider economy	Not separately estimated (included in GVA multiplier)
Deadweight	What would have happened anyway	Modular vs traditional counterfactual isolates incremental impact

Counterfactual	Baseline scenario without modular intervention	Traditional construction methods continue
Discounting	Converting future values to present value using 4% real discount rate	Applied to all multi-year cashflows
Fiscal Multiplier	Ratio of total economic output to initial government spending	Estimated at 1.2-1.3 for construction (international benchmark)

Two refinements merit explicit mention. First, while the CSO (2025) confirms a baseline construction GVA of approximately 38% of output, modular factory components exhibit higher productivity due to automation and digitalisation. A sensitivity range of 42-45% for the off-site manufacturing portion is therefore noted, though the core model retains the conservative 38% figure. Second, the CSO does not publish official Type II (induced) multipliers for construction; the 1.2-1.3 range reported here represents Type I (direct + indirect) effects only. Total economic impacts would be materially higher if induced spending through worker consumption were included, but this analysis adheres to verifiable CSO standards.

All calculations are based on the accompanying Excel model, which contains detailed assumptions, formulas, and scenario analysis. The model spans six sheets: Assumptions (input parameters), Housing Delivery (unit counts by type), Economic Impact (jobs, GVA, taxes), Rent Impact (garden unit scenarios), NPV Calculation (discounted cashflows), and Summary (key metrics dashboard).

3.2 Data Sources (April 2026)

Source	Key Data
ESRI Spring 2026 Commentary	36,248 completions in 2025; mid-30,000s forecast 2026-2027
Central Bank (Feb 2026)	Housing sector at "near full capacity"
Gov.ie – Delivering Homes (Nov 2025)	303,000 homes by 2030; €275bn NDP
Gov.ie – Modular proposal (21 April 2026)	Proposed planning exemption; pending ratification
Business Post (27 April 2026)	Rent-a-room disagreement delaying approval

CSO Construction Output Data	GVA per worker, employment figures
Budget 2026 / Finance Bill 2025	9% VAT for apartments; 125% corporate tax deduction (expires 31 December 2030)
Revenue Commissioners	Rent-a-room €14,000 cap
Residential Tenancies Act	Licensee vs tenant distinctions
SCSI 2026 Benchmarking Report	Construction costs: €397k national avg, €461k Dublin
Construction Industry (April 2026)	Labour shortages critical; wage inflation rising

Note: This analysis does not model activity beyond 2029. The 9% VAT rate for apartments and 125% corporation tax deduction are scheduled to expire on 31 December 2030. Projects completing after this date would revert to standard rates (13.5% VAT, standard deduction), reducing exchequer returns and developer viability. Readers should monitor potential extension of these measures

3.3 GVA and Tax Calculation Methodology (Per Unit)

Metric	Modular Apartment (High Density)	Modular House (Semi-D/Terrace)	Calculation Rationale
Gross Delivery Cost	€280,000	€320,000	SCSI 2026 benchmarks + 15-20% modular saving
VAT Rate	9.0% (Budget 2026)	13.5% (Standard)	Statutory rates
Net Construction Value	€256,881	€281,938	Gross ÷ (1 + VAT)
GVA per Unit (38% Multiplier)	€97,615	€107,136	CSO construction sector average
Direct VAT Revenue	€23,119	€38,062	Gross – Net

GVA Multiplier Justification: The 38% GVA multiplier is derived from CSO construction sector data showing that approximately 38% of gross output represents value added (labour, profit, capital consumption), with the remainder representing intermediate inputs (materials, subcontractor services). This is consistent with EU national accounts methodology (ESA 2010) and has been cross-referenced with UK ONS construction GVA ratios (typically 35-40%) for validation. Notably,

the CSO advises that its construction output data should be treated as indicative due to the sector's fragmented nature and high proportion of self-employed workers.

3.4 Counterfactual Construction Time and Cost Comparison

Metric	Traditional (Counterfactual)	Modular	Difference
Construction time per home	18-24 months	9 months	50-67% faster
Cost per home (house)	€350k-€450k	€280k-€320k	15-20% cheaper
Cost per home (apartment)	€325k-€400k	€250k-€300k	15-20% cheaper
On-site waste	15%	5%	66% less waste
Labour requirement (on-site)	100%	50%	50% reduction
Quality consistency	Variable (site-dependent)	Factory-controlled	Higher consistency

Economic Interpretation: The time saving of 9-15 months per home has material economic value:

- Earlier occupancy generates earlier rental/purchase revenue (estimated €15,000-25,000 per home in accelerated cashflow)
- Earlier exchequer returns (VAT, income tax) with positive time value of money (4% discount rate)
- Reduced construction finance costs for developers (interest savings of €5,000-10,000 per home)

AI Usage Disclosure: AI tools were used to assist with the drafting and editing of this report. All data, assumptions, and conclusions were verified by the author.

4. ASSUMPTIONS TABLE (APRIL 2026)

Parameter	2026 Value	Source / Rationale
Total Homes Target (2030)	303,000	Delivering Homes plan (Nov 2025)
2025 Completions (actual)	36,248	ESRI Spring 2026
2026 Forecast	36,500	ESRI – mid-30,000s plateau
2027 Forecast	38,000	ESRI conditional on VAT measures
2028 Forecast	39,500	Illustrative scenario (+4% growth)
2029 Forecast	41,000	Illustrative scenario (+4% growth)
Cost: Standard House	€350k – €450k	SCSI 2026 benchmarks; €397k national avg
Cost: Modular House	€280k – €320k	15-20% saving; absolute costs higher due to inflation
Cost: Modular Apartment	€265k – €315k	Base €250k-€300k +5% Jan 2026 fire safety regs
Cost: Back-garden modular unit	€50k – €120k	Varies by size, specification, utilities
VAT: Houses	13.5%	Standard rate
VAT: Apartment blocks (new)	9%	Budget 2026 – until 31 Dec 2030
Corporation Tax (large firms >€750m)	15%	OECD agreement effective 2024
Corp Tax Deduction (apartments 10+ units)	125% (max €50k/unit)	Finance Bill 2025; sunset 31 Dec 2030
Construction Employment	~175,000	CSO Q4 2025; labour shortages critical
Unemployment Rate	4.8 – 5.2%	ESRI forecast
Average Construction Salary	€58,500	CSO estimate + 2025 wage inflation
GVA Multiplier (Construction)	38%	CSO construction sector averages

Indirect Employment Multiplier	1.45 – 1.6	Off-site manufacturing creates stable upstream jobs
Discount Rate	4%	Irish Government Infrastructure Guidelines
Modular Planning Exemption	Proposed – pending Cabinet approval	Delayed due to rent-a-room disagreement
NSAI Modular Standards	Expected H1 2026	Four new certification standards expected
Single-Stage Approval (projects <€200m)	Active	Adopted Nov 2025; accelerates MMC projects
Rent-a-Room Scheme (garden units)	Under discussion	Key policy disagreement delaying approval
Rent-a-Room Annual Cap	€14,000	Revenue Commissioners

5. SCENARIO ANALYSIS

5.1 Policy Risk Scenarios

To assess the sensitivity of results to policy uncertainty, three scenarios are modelled:

Scenario	Modular Approval	Rent-a-Room for Garden Units	VAT (Apartments)	Corp Tax Deduction
Optimistic	Enacted H2 2026	Included	9% (extended beyond 2030)	125% (extended beyond 2030)
Base Case	Enacted 2027	Included	9% (until 2030)	125% (until 2030)
Pessimistic	Delayed beyond 2027	Excluded	13.5% (reverts)	Standard deduction

NPV Sensitivity to Policy Delay (6-month delay, 4% discount rate):

Project Value	Present Value Loss	% Loss
€100 million	~€2 million	2%
€1 billion	~€20 million	2%
€70 billion ((illustrative 5-year national modular investment from this analysis)	~€1.4 billion	2%

Economic Interpretation: Every 6 months of policy delay reduces the present value of the national modular housing programme by approximately €1.4 billion. This represents a significant economic cost of inaction that should be weighed against the political considerations delaying the planning exemption. This is an illustrative calculation using 4% discount rate; not directly modelled in Excel.

5.2 Modular Share Scenarios

Year	Low Modular Share	Base Modular Share	High Modular Share
2026	5%	8%	12%
2027	7%	10%	15%
2028	10%	12%	18%
2029	12%	15%	20%

Economic Interpretation: The modular share is constrained by (a) industry capacity to scale modular manufacturing (new factories typically take 18-24 months to reach full capacity), (b) labour availability (skilled modular assembly workers require training), and (c) policy certainty (developers will not invest in modular without stable planning rules). The Base Case assumes gradual scaling as modular factories come online and policy clarity improves following the expected 2027 enactment.

6. HOUSING DELIVERY FORECASTS (2025-2029)

6.1 Annual Completions (Base Case)

Year	Total Completions	Modular Share (Base)	Modular Homes	Traditional Homes
2025	36,248 (actual)	5%	1,812	34,436
2026	36,500	8%	2,920	33,580
2027	38,000	10%	3,800	34,200
2028	39,500	12%	4,740	34,760
2029	41,000	15%	6,150	34,850
Total	191,248	10% (avg)	19,422	171,826

Note: Completions are illustrative policy scenarios based on ESRI structural demand estimates (~44,000 per year). Actual CSO completions in 2025 were 30,330 units. This model demonstrates the economic impact of achieving government housing targets under modular adoption.

6.2 Housing Type Breakdown (Base Case, 2026)

Housing Type	Units	% of Total
Traditional Houses	20,148	55%
Traditional Apartments	13,432	37%
Modular Houses	1,752	5%
Modular Apartments	1,168	3%
Total	36,500	100%

Economic Insight: Apartments dominate Dublin completions (60-70% of new units), while houses dominate outside Dublin. The 9% VAT rate and 125% corp tax deduction specifically target apartments, aligning incentives with urban density needs. This split reflects the economic reality that apartments are the only viable solution for high-density urban areas where land prices exceed €1 million per hectare.

7. BACK-GARDEN MODULAR UNITS & RENT IMPLICATIONS

7.1 Typology and Planning Status

Modular Type	Size	Planning Status	Rent Implications
Large-scale modular developments	Full homes	Standard planning (or single-stage for state projects <€200m)	Standard market rent
Back-garden modular homes	≤45 sq m (with 25 sq m garden remaining)	Proposed exemption – currently delayed	Would fall under rent-a-room scheme (€14,000 tax-free)
Garden offices / cabins	Very small (<25 sq m)	Often exempt as outbuildings	Not for residential use

Policy Update (April 2026): The Government has proposed exempting modular garden units up to 45 square metres from planning permission, provided at least 25 square metres of garden space remains and the unit has side access to the property. These units qualify for the €14,000 tax-free Rent-a-Room Relief if rented privately. Larger units or those not meeting the criteria require a fast-tracked notification process.

7.2 Rent-A-Room Scenarios (Garden Units)

Scenario	Annual Rent (€)	Taxable Income	After-Tax Return (€)	Tenant Status
Included in rent-a-room	14,000	0	14,000	Licensee (fewer protections)
Excluded (standard rental)	20,000	20,000	~12,000	Tenant (full RTA protections)
Excluded (market rate)	25,000	25,000	~15,000	Tenant (full RTA protections)

Legal-Economic Insight: The distinction between licensee and tenant is not merely technical. Licensees under the rent-a-room scheme lack:

- Security of tenure (can be evicted with reasonable notice, no PRTB reference period)

- Rent pressure zone protections (no limits on rent increases)
- Access to Residential Tenancies Board dispute resolution

Policy Trade-Off:

- Including garden units under rent-a-room → Higher homeowner supply incentive (+€2,000-€3,000 annual after-tax return), but weaker tenant protections
- Excluding garden units → Stronger tenant protections, but reduced financial case for homeowners to build (after-tax return drops from €14,000 to ~€12,000)

The current policy disagreement centres on exactly this trade-off. Resolution is critical to unlocking the estimated 50,000-100,000 potential garden units that could be delivered on existing residential properties across Ireland.

8. ECONOMIC IMPACT RESULTS

8.1 Annual Economic Impact (2026)

Metric	Value	Calculation
Total investment	€13.4 billion	36,500 units × weighted avg cost
Direct construction jobs	22,883	Investment ÷ €58,500 avg salary × 0.1
Total GVA	€1,339 million	Investment × 0.38 GVA multiplier
Income tax revenue	€361 million	Direct jobs × €58,500 × 27% effective rate
VAT revenue (houses)	€1,174 million	Traditional houses × €397k × 13.5%
VAT revenue (apartments)	€422 million	Traditional apts × €280k × 9%
Corporation tax	€100 million	Estimated 15% on construction profits

8.2 Five-Year Cumulative Impact (2025-2029)

Metric	Total	Annual Average
Total homes delivered	191,248	38,250
Total modular homes	19,422	3,884
Total investment	€70.1 billion	€14.0 billion

Total direct jobs (person-years)	119,774	23,955
Total indirect jobs (person-years)	179,661	35,932
Total jobs supported	299,435	59,887
Total GVA	€7.0 billion	€1.4 billion
Total exchequer return	€10.8 billion	€2.2 billion

8.3 Net Present Value (5-Year, 4% Discount Rate)

Year	Net Benefit (€m)	Discount Factor	PV (€m)	Cumulative PV (€m)
2025	15,360	1.0000	15,360	15,360
2026	15,445	0.9615	14,851	30,211
2027	16,064	0.9246	14,852	45,063
2028	16,682	0.8890	14,830	59,893
2029	17,291	0.8548	14,780	74,673
NPV (€bn)				74.7

Economic Interpretation: The positive NPV of €74.7 billion indicates that modular housing delivery generates significant economic value over the 5-year period, substantially exceeding the cost of capital (4% discount rate). Every €1 invested in modular housing generates approximately €1.07 in present value terms, driven by the multiplier effects of construction spending and subsequent tax revenues.

9. REGIONAL DISTRIBUTION ANALYSIS

9.1 Share of Housing Output by Region (2025-2029)

Region	Share of Completions (%)	Share of Investment (%)	Key Drivers
Dublin	45-50%	50-55%	Highest demand, land constraints, apartment focus
Cork	12-15%	10-12%	Second city, MMC adoption
Galway	5-7%	4-6%	University-driven demand
Limerick	5-7%	4-6%	Urban centre
Rest of Leinster	15-18%	12-15%	Commuter belt
Rest of Munster + Connacht + Ulster	10-12%	8-10%	Rural areas

9.2 Regional GVA Impact (Base Case, 5-Year Total)

Region	GVA (€bn)	Share
Dublin	~€10-11	50-55%
Cork	~€2.5-3.0	12-15%
Galway	~€1.0-1.5	5-7%
Limerick	~€1.0-1.5	5-7%
Rest of Leinster	~€2.5-3.0	12-15%
Rest of country	~€2.0-2.5	10-12%

9.3 Regional Exchequer Return (Base Case, 5-Year Total)

Region	Exchequer Return (€bn)	Share
Dublin	~€3.5-4.0	50-55%
Cork	~€0.8-1.0	12-15%
Galway	~€0.3-0.5	5-7%

Limerick	~€0.3-0.5	5-7%
Rest of Leinster	~€0.8-1.0	12-15%
Rest of country	~€0.6-0.8	10-12%

Economic Insight: The Dublin concentration reflects economic fundamentals: highest land values, strongest demand, greatest viability for apartments with 9% VAT and 125% corp tax deduction. Regional balance requires targeted incentives for modular adoption outside Dublin, such as higher VAT reductions or capital grants for modular factories in regional centres.

10. COMPARISON TO ALTERNATIVES

10.1 Counterfactual Scenarios

Scenario	Description	5-Year Total Homes	GVA (€bn)	Exchequer Return (€bn)
Do Nothing	No intervention; traditional methods only	~175,000	~€18-19	~€5-6
Traditional Only	No modular; traditional methods	~180,000	~€19-20	~€6-7
Modular (Base Case)	Modular share 8-15%	~191,000	~€19-20	~€6-7
Modular (High)	Modular share 12-20%	~195,000	~€20-21	~€7-8

10.2 Incremental Impact of Modular

Metric	Incremental Value (5-Year)
Additional homes delivered	~11,000
Additional GVA	~€0.5-1.0 billion
Additional exchequer return	~€0.3-0.5 billion

Economic Insight: The incremental impact of modular is material but not transformative. Modular's primary value is not increasing total volume (constrained by labour availability) but delivering the same volume faster and cheaper, with positive time value benefits. The time saving of 9-12 months per home generates earlier occupancy, earlier tax receipts, and reduced construction finance costs – benefits that are partially captured in the NPV calculation (4% discount rate) but not fully reflected in volume-based metrics.

11. LIMITATIONS AND CAVEATS

11.1 Data Limitations

Limitation	Impact	Mitigation
No recent Irish indirect multiplier data	Cannot quantify supply chain effects with Irish-specific accuracy	Use international benchmarks (1.2-1.3x) with explicit caveat
CSO input-output tables last updated 2000	Construction multipliers severely outdated	Rely on GVA per worker methodology instead
Labour shortage dynamics not modelled	Job estimates assume labour availability – optimistic	Explicit caveat in all job projections
Forecasts beyond 2027 highly uncertain	2028-2029 figures illustrative	Label clearly as "scenarios" not "forecasts"
Modular approval timing uncertain	Base Case assumes 2027 enactment	Three scenario analysis (Optimistic/Base/Pessimistic)

11.2 Policy Uncertainty

Uncertainty	Current Status	Implication
Modular planning exemption	Proposed – pending Cabinet approval; delayed due to rent-a-room disagreement	Delays project starts; reduces NPV by ~2% per 6 months

VAT 9% rate for apartments	Expires 31 December 2030	Cliff-edge uncertainty for projects completing after 2030
125% corporation tax deduction	Expires 31 December 2030	Cliff-edge uncertainty for apartment blocks of 10+ units
Rent-a-room applicability to garden units	Under discussion	Affects homeowner supply incentives for back-garden units

11.3 Structural Constraints

Constraint	Severity	Implication
Construction labour shortage	Critical – "tens of thousands of additional workers needed"	Binding constraint on housing delivery regardless of modular adoption
Modular manufacturing capacity	Limited – scaling requires time and investment	Gradual modular share increase (5% → 15% over 5 years)
Land availability (especially Dublin)	Constrained	Favours apartments over houses
Planning system capacity	Strained	Single-stage approval for projects <€200m helps but is not a panacea

11.4 Completions Caveat

As noted in the Summary sheet of the accompanying Excel model:

This analysis assumes completions increase from 36,000 in 2026 to 41,000 in 2029, consistent with ESRI structural demand estimates. Actual CSO completions in 2025 were 30,330 units. The model illustrates the economic impact of achieving policy targets under modular adoption.

12. KEY FINDINGS & RECOMMENDATIONS

12.1 Key Findings

Finding	Evidence	Implication
Modular delivers significant efficiency gains	50% faster, 15-20% cheaper, 66% less waste	Accelerates delivery, reduces costs
Policy uncertainty is the critical constraint	Planning exemption delayed due to rent-a-room disagreement	Each month of delay reduces NPV by ~2% (€230 million per month nationally)
VAT differential (9% vs 13.5%) favours apartments	Budget 2026 measure	Improves apartment viability significantly
125% corp tax deduction for apartments (10+ units)	Finance Bill 2025	Effective subsidy of €10k-12.5k per unit
Garden units have different economics	Rent-a-room (€14k tax-free) vs standard rental (~€12k after tax)	Material impact on homeowner supply incentives
Labour shortages remain severe	ESRI / Central Bank warnings: tens of thousands of additional workers needed	Binding constraint regardless of modular adoption
Dublin dominates economic impact	50-55% of GVA and exchequer returns	Regional balance requires targeted incentives

12.2 Recommendations

Recommendation	Rationale	Responsible Party
Resolve rent-a-room disagreement	Unlocks planning exemption for garden modular units	Cabinet / Coalition parties
Expedite modular approval ratification	Proposed April 2026 regulations should be prioritised	Minister for Housing
Extend VAT measures beyond 2030	Provides long-term certainty for apartment developers	Minister for Finance

Extend corp tax deduction beyond 2030	Incentivises large-scale modular apartment blocks	Minister for Finance
Invest in construction labour and skills	Address critical shortage via apprenticeship expansion, targeted immigration	Minister for Further Education, Minister for Enterprise
Update CSO input-output tables	No recent indirect multiplier data available for Ireland	CSO / Minister for Public Expenditure
Develop regional modular incentives	Balance Dublin concentration with regional delivery	NTA / Local authorities

13. CONCLUSION

This Economic Impact Assessment confirms that modular construction offers significant efficiency gains for Ireland's housing delivery: 50% faster completion, 15-20% lower costs, and 66% less on-site waste.

However, three conditions must be met for these gains to be realised:

Condition	Current Status	Action Required
1. Policy certainty	Modular planning exemption proposed but delayed (rent-a-room disagreement)	Cabinet resolution needed
2. Labour availability	Severe shortage; tens of thousands of additional workers needed	Apprenticeship expansion, targeted immigration pathways, and accelerated integration of international protection applicants into construction employment.
3. Manufacturing capacity	Scaling requires time and investment	Industry development support

Economic Bottom Line:

Metric	Value
5-year NPV (Base Case)	€74.7 billion
NPV loss per 6-month policy delay	~€1.4 billion (based on illustrative €70bn national programme)
Incremental homes from modular (Base vs Traditional)	~11,000 over 5 years
Total jobs supported	299,435 person-years
Total exchequer return	€10.8 billion

Final Verdict: Modular construction is a necessary but not sufficient condition for meeting Ireland's housing targets. Without policy certainty (resolution of the rent-a-room disagreement) and labour market interventions (apprenticeship expansion, targeted immigration, and accelerated integration of international protection applicants into construction employment), the efficiency gains will remain only partially realised. The economic case for rapid resolution is clear: each month of delay costs approximately €230 million in present value terms, based on illustrative national modular investment of €70 billion over 5 years (from this analysis).

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